

Vision-Guided Expressive Motion for SOFIA: Technology Analysis and Preliminary System Design

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Abstract—Interactive robotic surfaces require more than visually coherent animation: perception, motion generation, communication, and electromechanical limits must agree. This report analyzes SOFIA, a demonstrator containing sixteen two-axis hexagonal units, and develops a preliminary proposal for vision-guided tracking and persistent gesture-triggered motion. Following the Discover and Learn stages of D-Lab, the study combines a review of the provided architecture and course material, a comparison with relevant literature, manufacturer-document checks, analytical modeling, and reproducible software microtests. The selected opportunity is a modular interaction pipeline that recognizes five postures or gestures, including three dynamic gestures, and translates the selected user’s intent into bounded motion. A two-hand heart gesture triggers a repeating outward wave. The proposed architecture retains the ESP32-C3, two PCA9685 controllers, and thirty-two servos while separating perception on a host computer from local command supervision. Independent numerical checks cover eight hundred unit-target configurations, singular cases, reference smoothing, and a conservative panel-clearance screen. The analysis identifies unresolved power, torque, calibration, collision, communication, and perception risks and assigns explicit acceptance tests. These outputs establish a traceable technical proposal; physical performance and gesture-recognition accuracy remain to be measured during subsequent integration.

Index Terms—SOFIA, kinetic surfaces, human–robot interaction, inverse kinematics, gesture recognition, D-Lab.

I. INTRODUCTION

INTERACTIVE kinetic surfaces turn digital information into physical motion, making them useful for expressive installations and for teaching the connection between perception, geometry, electronics, and control. A convincing response, however, must remain interpretable when camera observations fluctuate, several people appear, or an actuator cannot follow the requested motion. A successful animation alone does not establish electrical compatibility, mechanical clearance, or repeatable interaction.

The SOFIA demonstrator described in the supplied technical dossier consists of a honeycomb arrangement of sixteen hexagonal units, each with two rotational degrees of freedom and two MG90S servos. One ESP32-C3 board and two PCA9685 devices provide control for a complete sixteen-unit assembly; additional assemblies can communicate over RS-485/Modbus [1]. In this report, *unit* means one two-axis mechanism

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and *panel* means the complete sixteen-unit assembly. This distinction avoids confusing the course’s occasional use of “module” for both scales.

The formal problem is to convert uncertain visual observations into coordinated, bounded orientation commands for this panel while preserving the intended interaction state. The guiding question is: *How can a camera-based interface provide persistent, understandable collective motion on SOFIA while respecting its kinematic and electrical constraints?* The suggested product is an indoor educational installation with manual, automatic-target, body-tracking, and gesture modes, plus a graphical interface showing camera landmarks, the active interaction, and faults. The course specifies five postures, multiple visible people, wireless communication, and at least three dynamic postures [2].

The technical objectives are to explain the reference architecture, select a defensible perception-to-motion approach, define an explicit kinematic and communication interface, and identify tests that decide whether the proposal can proceed to integration. Three research questions organize these objectives: RQ1, can the geometric model generate feasible, continuous orientation commands? RQ2, can temporal gesture rules preserve an expressive state without unwanted switching? RQ3, can the electronics and communication deliver those commands within measured power and timing limits? This Discover–Learn report closes the analytical and documentary tasks and defines the remaining empirical work; it does not claim a validated physical installation [3].

A. Reference System, Actors, and Functional Decomposition

The documented information path is camera/user input to a PC, wireless commands to a master controller, and local PWM generation for the servos; an internal serial bus supports expansion [1], [2]. Table I translates that architecture into functions and interfaces. The basic geometric model directs the output-link axis toward a target. It does not give the panel arbitrary three-axis orientation or make the panel center coincide with the target.

The supplied Assignment 1 report documents a MATLAB model with staggered rows of 3, 4, 5, and 4 units, 120 mm pitch, and an ideal joint interval of $[-90^\circ, 90^\circ]$. It reports a maximum direction error near 1.71×10^{-6} degrees over 50 samples and sixteen units [4]. These are numerical consistency results, not measured servo accuracy. Both uploaded copies are byte-identical and constitute one source. They contain simulation images, although their captions retain wording about reserved

TABLE I
FUNCTIONAL DECOMPOSITION AND STAKEHOLDER NEEDS.

Subsystem	Function and interface	Stakeholder need
Perception	Camera frames to person/hand landmarks	Visitors: understandable interaction
Interaction	Landmarks to selected person, gesture, and mode	Operator: visible state and stop control
Motion	Target or pattern to 32 angle references	Students: explainable geometry and limits
Embedded control	Commands to calibrated PWM; local watchdog	Integrators: deterministic response
Mechanics/power	Servos to panel orientation; separate supply paths	Maintainers: accessible, protected hardware
Evidence	Events and settings to time-stamped records	Instructors: reproducible decisions and tests

frames. The supplied component assessment also leaves servo current, torque margin, pulse endpoints, and environmental requirements unresolved [5].

B. Related Work and Technological Benchmark

Follmer et al. demonstrate physical interaction through the inFORM shape display, using 900 actuated pins and depth sensing [6]. Its translation-based surface shows that distributed actuation can support expressive interaction, but its hardware and direct physical manipulation differ from SOFIA’s rotating hexagons. Consequently, its motion rates and force capabilities are not transferable SOFIA specifications.

Zhang et al. combine palm detection with hand-landmark estimation from an RGB camera [7]. Their work supports using geometric hand features instead of interpreting raw pixels for every gesture. Landmark estimation alone does not define a heart gesture, temporal intent, user ownership, or actuator commands. Cao et al. address multi-person body-pose association through part affinity fields in OpenPose [8]. This is relevant when several users appear together, but it likewise leaves SOFIA’s interaction policy and mechanics to the application.

For motion modeling, Lizarraga et al. study inverse kinematics and angular continuity in hexapod locomotion [9]. Their robot differs from SOFIA, but the attention to angle branches motivates explicit branch and singularity handling. Their later planar-workspace study [10] and the parallelogram framework of Luque-Vega et al. [11] distinguish admissible geometry from joint-angle limits. SOFIA adopts that distinction; the specialized boundary equations for those robots cannot be reused as collision proofs for spatial panels.

Table II compares relevant capabilities rather than combining unrelated accuracy or frame-rate results. The opportunity is not the invention of hand tracking or inverse kinematics. It is their traceable integration for SOFIA: an explicit five-gesture vocabulary, persistent interaction states, bounded collective patterns, and local supervision. No superiority over the reviewed systems is claimed.

C. Opportunity Map and Selected Scope

The source review identifies four linked opportunities. First, an explicit visual explanation of landmarks and active states can make the installation easier to understand and debug. Second, temporal evidence and a latched gesture state can prevent a repeating motion from stopping whenever the hands leave a recognized pose. Third, model-based workspace checks can expose commands that are mathematically defined but mechanically unacceptable. Fourth, calibration and timestamped records can connect simulation assumptions to hardware evidence.

The selected opportunity combines the first two as the user-facing function and treats the latter two as engineering requirements. The heart gesture produces an outward traveling wave in the honeycomb arrangement. Other gestures select distinct bounded patterns. Detection continues while the selected pattern repeats; a new confirmed gesture changes the pattern. Operator stop and faults take priority. The available project history motivates this interaction, but no recognition rate or physical success is inferred from that history. Table III makes the scope testable.

II. MATERIALS AND METHODS

A. Study Design, Evidence, and Decision Process

The Discover stage used the supplied architecture, CAD drawing sheets, kinematic report, component review, computational-systems brief, and D-Lab checklist. The Learn stage compared relevant research with those requirements, checked critical manufacturer specifications, derived design budgets, and executed independent software microtests. Course material on filtering and Modbus informed the reference-conditioning and bus analyses [12], [13]. The supplied residential-water classification article is useful as a reporting example but is not a direct technological benchmark for this installation.

Evidence is labeled as *documented* in supplied material, *calculated* from stated assumptions, *software-tested* with the included scripts, or *proposed* for physical evaluation. No laboratory measurements, user trials, camera dataset, servo calibration file, or original MATLAB source were supplied. The numerical reconstruction therefore defines its own reproducible coordinate origin and sampling convention. It is not an exact reproduction of the unavailable program. The source register and review log in the supplementary package preserve document names, pages, decisions, and unresolved items.

B. Materials, Component Selection, and Preliminary BOM

The baseline retains the reference servo topology because it matches the provided mechanism and already provides exactly 32 command channels. Direct MCU PWM would change resource allocation and wiring without resolving the primary perception and calibration questions. A different actuator remains an option if the measured torque or thermal margin is insufficient. Table IV is a preliminary, cost-free assembly BOM; unverified vendor details and CAD-dependent quantities are not invented.

TABLE II
LITERATURE AND REFERENCE-SYSTEM COMPARISON. PROPOSED CAPABILITIES ARE DESIGN INTENTIONS, NOT MEASURED RESULTS.

Source/system	Method or documented capability	Useful evidence for SOFIA	Boundary of applicability
SOFIA dossier and Assignment 1 [1], [4]	16 two-axis units; 32 PWM outputs; analytical target tracking	Known topology and a numerical forward/inverse model	Integrated gesture accuracy, physical clearance, and power margin are not established
inFORM [6]	900 translating pins; depth-camera interaction	Precedent for coordinated physical displays	Different mechanism, sensing, and interaction task
MediaPipe Hands [7]	Palm detector and 21-point hand model	Candidate observations for rule-based hand gestures	No SOFIA gesture vocabulary or motion controller
OpenPose [8]	Bottom-up multi-person pose association	Alternative for crowded body-pose scenes	Deployment speed and hand quality require local evaluation
Kinematic/workspace studies [9]–[11]	Angle handling and geometry-dependent feasible regions	Need to check branches, singularities, and collisions separately	Different robot chains; constraints must be rederived
This proposal	RGB perception; five gestures; latched patterns; bounded references	Testable requirements and software-only microtests	Hardware and human-interaction acceptance remain open

TABLE III
PROPOSED REQUIREMENTS AND ACCEPTANCE CRITERIA. NUMERICAL TARGETS ARE INITIAL PROJECT TARGETS UNLESS IDENTIFIED AS DEVICE LIMITS.

ID	Requirement	Acceptance criterion and measurement	Current evidence
R1	Explainable 16-unit orientation	Forward/inverse direction error $< 10^{-4}$ degrees for reachable synthetic targets; explicit zero-range, pole, and unreachable handling	Software microtests; no physical accuracy claim
R2	Five gestures, three dynamic; multiple people	Detect all five classes with 1, 2, and 3 visible participants; event macro-F1 ≥ 0.90 in each condition; at most one false activation per minute of negative activity	Proposed human-interaction tests
R3	Persistent expressive state	Heart wave continues for 10 s after hands leave the pose; confirmed alternative or Stop changes state; no phase reset on repeated heart detections	Proposed state-machine test
R4	Responsive interface and commands	Display rate ≥ 20 frames/s; 95th-percentile confirmed-event-to-ACK latency ≤ 250 ms; recognition and physical-response delays reported separately	Proposed timing measurements
R5	Bounded, noninterfering motion	Calibrated angle bounds; command slew $\leq 90^\circ/\text{s}$; at least 2 mm modeled and measured clearance along transitions	Rate microtest and panel-only screen; CAD/hardware pending
R6	Compatible logic and power	3.3 V logic within chip limits; 5 V servo rail within verified unit limits; measured current below the weakest derated path rating	Datasheet review; loaded measurements pending
R7	Communication-fault containment	Missing valid heartbeat for 500 ms enters Hold; invalid/stale/out-of-range commands do not move servos; deliberate operator recovery	Proposed fault-injection tests
R8	Calibration and physical pointing	Independent measured steady direction error $\leq 5^\circ$ over a validated target grid; record hysteresis and worst case	Initial target; physical tests pending
R9	Reproducible evidence	Each test records configuration, units, timestamps, raw observations, and verdict; versioned manuscript and team log	Local package prepared; team approval and GitHub pending

The electronics review resolves a naming hazard: the dossier calls its high-current 5 V rail V_{DD} and its electronics 5 V rail V_{CC} , whereas an IC datasheet may use V_{DD} for a 3.3 V logic supply. Here the unambiguous labels are `SERVO_5V`, `ELEC_5V`, and `LOGIC_3V3`. They must not be connected solely because their source documents use similar names.

The ESP32-C3 silicon operates from 3.0–3.6 V; its supply must support at least 0.5 A, and GPIO2, GPIO8, and GPIO9 require boot-configuration care [14, Secs. 3 and 5.2]. The Super Mini regulator and board pinout must be inspected separately. The PCA9685 logic supply accepts 2.3–5.5 V, so 3.3 V logic and pull-ups are selected. Addresses 0x40 and 0x41 distinguish the two ICs; unused group-address responses should be disabled during address testing [15, Secs. 7.1, 7.3.5; Table 14]. Its PWM outputs carry command signals, not motor supply current. Servo power travels through the distribution wiring and board power traces.

The official MG90S page specifies 1.8 kgf·cm stall torque and 0.10 s per 60° at 4.8 V, a mass of 13.4 g, and a 22.8×12.2

$\times 28.5$ mm envelope [16]. It does not establish continuous torque, current peaks, or calibrated pulse endpoints. Its higher-voltage torque entry differs from the course’s 6.0 V attribution; therefore this design uses only the explicitly specified 4.8 V torque datum for screening. The reference’s nominal 5 V servo rail remains subject to verification on the actual units. Reception of a 3.3 V PWM signal is also an empirical compatibility check; a suitable unidirectional buffer is required if the received HIGH level is inadequate.

The course Modbus example uses a MAX485 module powered at 5 V [13]. Its receiver output must not be assumed compatible with an ESP32-C3 input. A 3.3 V SN65HVD72-class interface is proposed [17]; retaining an existing 5 V board requires verified level translation. This is a deliberate interface decision, not a claim that the reference PCB has already been modified.

TABLE IV
PRELIMINARY BILL OF MATERIALS FOR ONE COMPLETE SIXTEEN-UNIT PANEL AND ITS HOST.

Qty.	Component or assembly	Identification and function	Selection status
32	MG90S positional microservo	Tower Pro; two axes per unit	Reference part; actual supplier, pulse range, current, and provenance to verify
2	PCA9685 PWM controller board	NXP IC; 16 channels, 12-bit PWM per IC	Retained; board vendor and power traces to inspect
1	ESP32-C3 Super Mini board	Espressif IC; local command supervision	Retained; exact board regulator, revision, and pinout to verify
1	RS-485 transceiver per panel	TI SN65HVD72 candidate; 3.3 V interface	Proposed compatible alternative; requires wiring/footprint verification
2	Five-contact inter-panel connectors	JST-VH interface in reference dossier	Exact housing, contacts, wire gauge, and rating to confirm
1	RGB status indicator assembly	LED with required current-limiting components	Documented function; final circuit to specify
16	Hexagonal moving-platform assemblies	Platform, serial-axis brackets, servo horns	Fabricated; material, mass, fasteners, and tolerances from final CAD BOM
1 set	Mounting frame and service enclosure	Fixed support and protected electronics	Fabricated; stiffness and access to verify
1 set	Regulated power and distribution	5 V actuator supply; separate electronics branch; 3.3 V regulation	Current rating selected only after transient measurements
1 set	Protection and interconnects	Branch fuses, switch, wiring, decoupling, bus termination as required	Values depend on verified layout and load
1	RGB camera and rigid mount	Host-connected camera; initial 640×480 at 30 frames/s	Proposed acquisition setting; exact camera and calibration pending
1	Windows host and GUI	Perception, interaction, simulation, logging	Existing host to inventory; record CPU/GPU, OS, and dependency versions

C. Preliminary Architecture and Technology Alternatives

Figure 1 defines the system boundary and separates data, motion commands, and power. Perception runs on the PC; the embedded controller performs command checks, calibration mapping, PWM updates, and fault supervision. The host receives acknowledgments and status, but standard servo PWM does not return the measured shaft angle. External orientation sensing is therefore required for physical tracking tests.

The preliminary host choice is a Windows Python application with OpenCV acquisition, MediaPipe body/hand landmarks, and a nonblocking GUI. The project’s browser prototype remains useful for interaction exploration; its visual behavior is not evidence that the hardware link is complete. MediaPipe’s current Pose Landmarker exposes a configurable maximum number of poses, while Hand Landmarker supplies 21 landmarks per detected hand [18], [19]. A three-person configuration must be benchmarked on the actual host. Hands can be processed within each tracked person’s region and assigned by proximity to that person’s wrists; ambiguous assignments are rejected. Persistent person identities are application logic, not assumed to be guaranteed by landmark indices.

OpenPose is the principal alternative for multi-person body association [8]. It is retained as a comparison if the selected pipeline cannot meet R2 or R4. Simple contour rules offer low implementation complexity but are sensitive to background and lack the anatomical features needed for the selected vocabulary. A depth camera would reduce some geometric ambiguity but changes the sensing hardware. Thus RGB landmarks are selected provisionally for reuse of the available camera, inspectable features, and deployment simplicity; no unmeasured speed ranking is asserted.

Wi-Fi is proposed for the PC-to-controller link. Bluetooth Low Energy is an alternative, but a Bluetooth Classic serial-port profile is not assumed for the ESP32-C3 [14]. The logical command record contains protocol version, session

identifier, sequence number, target panel, mode, 32 calibrated angle references or pattern parameters, and validity period. The receiver checks length, finite numeric values, bounds, and sequence freshness before acknowledging acceptance. Rejected messages return an error code. Acknowledgment means command acceptance, not confirmed mechanical arrival. Reconnection starts a new session with an explicit operator resume; stale queued commands are discarded.

For several panels, only one node initiates RS-485 transactions. Modbus address, CRC, timeout, and transmit-enable handling follow the course exercise, with a new register map needed for 32-axis commands [13]. A bus timing calculation illustrates the scaling constraint. At an assumed 115200 bit/s and 11 bits per character, a 73-byte write request and 8-byte reply consume

$$t_{\text{wire}} = \frac{11(73 + 8)}{115200} = 7.73 \text{ ms.} \quad (1)$$

This excludes required frame gaps, processing, and retries. A 30 Hz cycle cannot be assigned to an arbitrary number of panels; measure the complete polling time before expansion. The 73-byte request assumes 32 two-byte registers and nine overhead bytes; additional metadata increases it.

D. Geometric Model, Workspace, and Clearance

The model follows the supplied kinematic notes [20, pp. 9–12]. For unit i , let $\mathbf{b}_i$ be its base position, R_i its fixed mounting rotation, $h = 60$ mm the fixed offset, and $l = 100$ mm the modeled output-link length. These lengths are inherited *simulation parameters*; the drawing dimensions do not by themselves prove identical physical joint offsets. Axis locations, signs, and lengths must be registered to the assembled CAD before hardware use. With $\mathbf{e}_z = [0, 0, 1]^T$,

$$\mathbf{o}_{2i} = \mathbf{b}_i + R_i h \mathbf{e}_z, \quad (2)$$

$$\mathbf{o}_{3i} = \mathbf{o}_{2i} + R_i R_x(q_{1i}) R_y(q_{2i}) l \mathbf{e}_z. \quad (3)$$

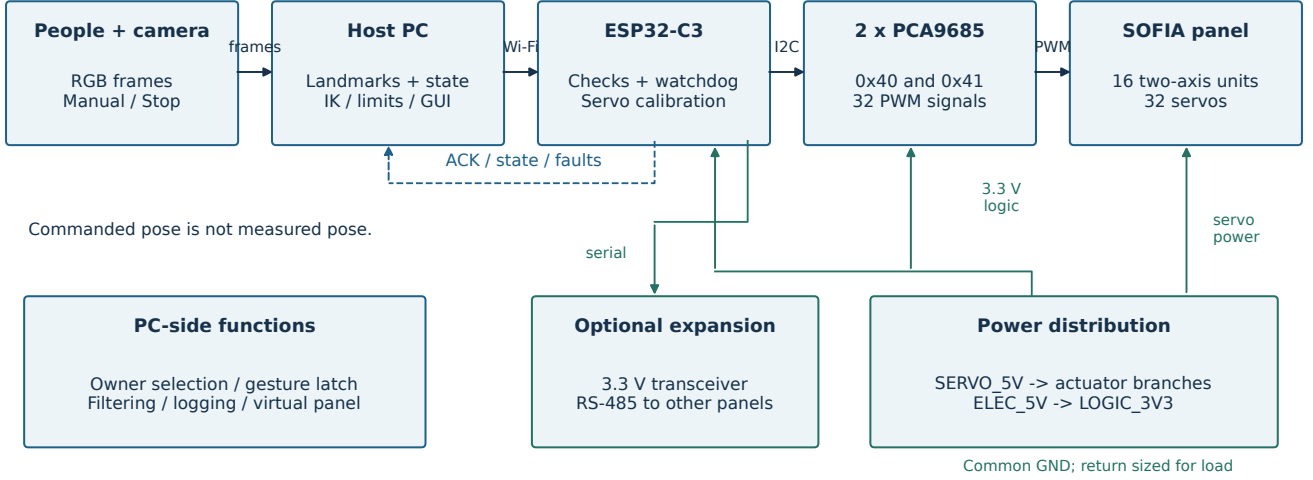


Fig. 1. Proposed architecture for one panel, with optional inter-panel expansion. Solid arrows carry observations or commands; the dashed return carries status, not measured joint position. Actuator and logic power are separated, with an engineered common reference.

The rotation order represents the serial mounting. Reversing it changes the mechanism. For a target $\mathbf{p}$ in the global frame, define

$$\mathbf{u}_i = \frac{R_i^T(\mathbf{p} - \mathbf{o}_{2i})}{\|\mathbf{p} - \mathbf{o}_{2i}\|} = [u_x, u_y, u_z]^T. \quad (4)$$

The local output direction and primary inverse solution are

$$\mathbf{n}_i = [\sin q_{2i}, -\cos q_{2i} \sin q_{1i}, \cos q_{1i} \cos q_{2i}]^T, \quad (5)$$

$$q_{2i} = \arcsin(\text{clip}(u_x, -1, 1)), \quad q_{1i} = \text{atan2}(-u_y, u_z). \quad (6)$$

The alternative branch is $(q_1 + \pi, \pi - q_2)$, wrapped to $[-\pi, \pi)$. Among feasible branches, select the one closest to the previous command. If neither is feasible, hold the previous accepted pose and mark the target unreachable. This explicit policy replaces the older report's clipping fallback; clipping is not claimed to minimize pointing error or guarantee clearance.

Two exceptional cases are distinct. At target distance below 10^{-8} mm the synthetic model retains both angles because direction is undefined. When $\sqrt{u_y^2 + u_z^2} < 10^{-6}$, q_1 is indeterminate: preserve its last value and accept $q_2 = \pm\pi/2$ only if allowed; otherwise hold. The second condition is the angular singularity described in the course. Hardware operation should use a margin from singular poses and calibrated joint limits, rather than assuming every MG90S provides the model's full $\pm 90^\circ$ interval.

Figure 2 specifies the reproducible lattice and pointing convention. For row $r = 0, 1, 2, 3$ containing $n_r = (3, 4, 5, 4)$ units,

$$x_{rj} = s \left(j - \frac{n_r - 1}{2} \right), \quad y_r = (\bar{r} - r) \frac{\sqrt{3}s}{2}, \quad \bar{r} = \frac{26}{16}, \quad (7)$$

where $j = 0, \dots, n_r - 1$, $s = 120$ mm, and all base z coordinates are zero. Numbering proceeds left to right and then top to bottom. $R_i = I$ in the microtests. The centroid lies between units, so a center-outward wave means increasing

distance from this centroid, not motion from a nonexistent central hexagon.

Clearance is a separate condition. With a preliminary circumsphere radius $a = 98.62/2 = 49.31$ mm around each moving platform center, a conservative panel-only screen is

$$\|\mathbf{o}_{3i} - \mathbf{o}_{3j}\| \geq 2a + g, \quad i \neq j, \quad g = 2 \text{ mm}. \quad (8)$$

Passing this screen excludes overlap of the assumed zero-thickness panel envelopes. It does not check the brackets, servo bodies, cables, frame, or dimensional uncertainty. Final checks require the actual swept solids and an inflated envelope accounting for thickness and tolerances. Both final poses and their interpolated transitions must be checked. Neutral pitch alone therefore cannot guarantee safe motion [10], [11].

E. Perception, Gesture Rules, and Persistent Motion

Body tracking uses a torso or shoulder-center image point; full-body mode additionally requires visible lower-body landmarks, whereas upper-body mode excludes that requirement. Gesture mode overlays both hands' detected landmarks and connections, not only the body skeleton. Image coordinates are normalized before distance thresholds are applied. Shoulder width normalizes body motion, while wrist-to-middle-finger-base distance normalizes a hand. Degenerate scales and occluded required points invalidate the observation.

An RGB landmark's relative depth must not be treated as a calibrated camera-to-panel distance [19]. For initial tracking, map the image point into a calibrated virtual plane of known panel-frame position. Given an undistorted pixel $\tilde{\mathbf{z}} = [u, v, 1]^T$, camera matrix K , rotation R_{0c} , and center $\mathbf{c}$, intersect its ray with $\mathbf{n}_\pi^T \mathbf{p} = d_\pi$:

$$\mathbf{p} = \mathbf{c} + \lambda R_{0c} K^{-1} \tilde{\mathbf{z}}, \quad \lambda = \frac{d_\pi - \mathbf{n}_\pi^T \mathbf{c}}{\mathbf{n}_\pi^T R_{0c} K^{-1} \tilde{\mathbf{z}}}. \quad (9)$$

Reject nearly parallel rays and nonpositive λ . The result is a virtual target on that plane, not a measured 3D body location.

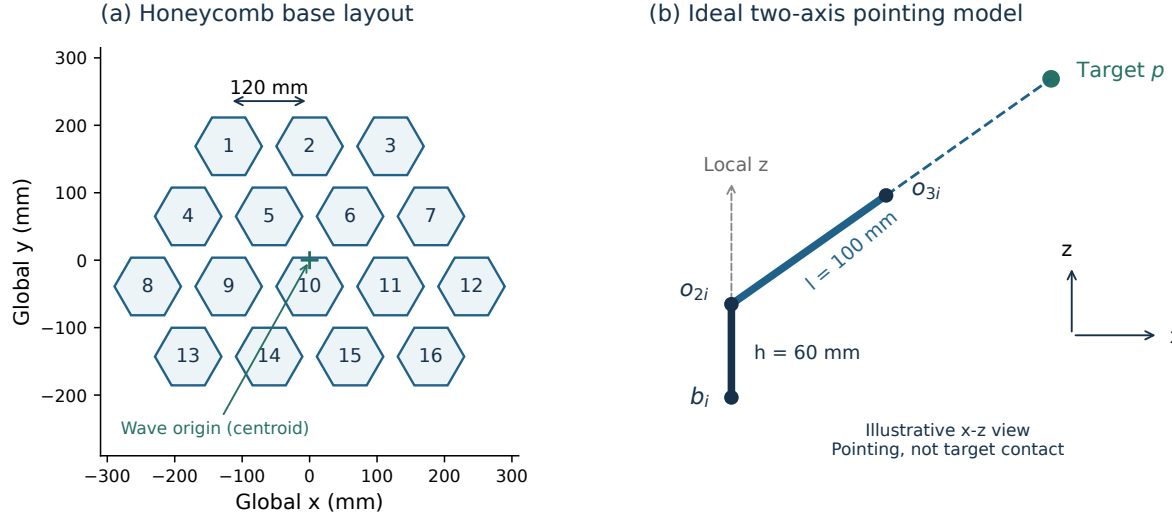


Fig. 2. Analytical geometry, not a manufacturing drawing. Left: sixteen base locations, unit IDs, and 120 mm nearest-neighbor pitch. Hexagon outlines use the nominal 98.62 mm diameter only for illustration. Right: the output axis points toward the target while its endpoint remains one modeled link length from the pivot.

Extrinsic calibration uses visible reference points tied to the frame.

Table V defines the proposed five-class vocabulary. The first three classes require ordered observations over time; a static pose followed by a dwell interval is not counted as a dynamic gesture. Initial thresholds are tunable design parameters, not learned performance claims. A valid observation initially requires relevant body visibility scores of at least 0.6, successful hand detection, and plausible hand-to-body association. The hand detector’s confidence is not interpreted as a per-joint visibility measurement.

For the heart rule, “inward-facing” means each index base lies outward of its own index tip relative to the body center; index bends must lie between 60° and 150° . These checks reject several touching-hand configurations but remain a heuristic requiring negative examples. A two-hand association must belong to the same tracked person. Dynamic detectors require at least five valid observations spanning the specified motion and reset after a 0.2 s observation gap. A 0.5 s refractory interval prevents repeated events from one completed motion.

The initial ownership rule selects the valid torso closest to the image center, retains that track while visible, and waits 1 s before replacing a lost owner. Ambiguous reassociation produces no new gesture. All visible people may be analyzed, but only the owner issues motion commands. Gesture features are computed independently for each track. Conflicting gestures within an owner’s window do not change state until one rule is unambiguous.

The behavior state records the mode, owner identity, active pattern, and pattern start time. A confirmed new gesture changes the pattern once. Repeated detections of the current gesture do not reset time. Absence of a recognized hand pose does not clear a latched pattern, so the heart wave continues with a healthy host heartbeat. Tracking mode instead holds the last target when its owner is lost. Stop, communication timeout, or an invalid motion envelope overrides either mode; continuation

requires deliberate recovery.

For the heart wave, let $\rho_i = \sqrt{x_i^2 + y_i^2}$ and $\phi_i = \text{atan2}(y_i, x_i)$. A proposed small-angle radial command is

$$w_i(t) = \sin\left(2\pi f(t - t_0) - \frac{2\pi\rho_i}{\Lambda}\right), \quad (10)$$

$$q_{1i}^*(t) = -A \sin \phi_i w_i(t), \quad q_{2i}^*(t) = A \cos \phi_i w_i(t). \quad (11)$$

Here $A = 5^\circ$, $f = 0.5$ Hz, and $\Lambda = 240$ mm are preliminary software-test values. Constant phase travels outward at $f\Lambda = 120$ mm/s; this is wave propagation across the layout, not platform translation speed. Mode changes blend from the last commanded pose and pass through the same limits and collision screen as tracking. Other patterns use the same initial amplitude and rate envelope.

F. Reference Conditioning and Error Budget

Following the course’s open-loop filtering discussion [12], the initial reference conditioner is an exponential moving average (EMA) followed by a slew limiter:

$$\bar{q}_k = \alpha q_k^* + (1 - \alpha)\bar{q}_{k-1}, \quad \alpha = 1 - e^{-\Delta t/\tau}, \quad (12)$$

$$q_k^c = q_{k-1}^c + \text{clip}(\bar{q}_k - q_{k-1}^c, -v_{\max}\Delta t, v_{\max}\Delta t). \quad (13)$$

The initial values are $\tau = 0.10$ s and $v_{\max} = 90^\circ/\text{s}$. At 30 Hz, $\alpha = 0.2835$ and the maximum command change is 3° . These states shape commands; they are not feedback measurements of actual servo position. Reset or initialize filter states to the current accepted command at a controlled mode transition. Final angle and geometry checks occur after conditioning, because independent filtering can alter a coordinated path.

The direction error in the ideal model is

$$e_{\theta i} = \cos^{-1}\left[\text{clip}((R_i \mathbf{n}_i)^T \hat{\mathbf{d}}_i, -1, 1)\right], \quad \hat{\mathbf{d}}_i = \frac{\mathbf{p} - \mathbf{o}_{2i}}{\|\mathbf{p} - \mathbf{o}_{2i}\|}. \quad (14)$$

Near perfect alignment, an $\text{atan2}(\|\mathbf{n} \times \mathbf{u}\|, \mathbf{n}^T \mathbf{u})$ evaluation is numerically better conditioned. Neither metric includes vision

TABLE V
PROPOSED GESTURE VOCABULARY AND REACTIONS. THRESHOLDS MUST BE TUNED ON DEVELOPMENT DATA BEFORE HELD-OUT EVALUATION.

Gesture	Type	Initial geometric/temporal rule	Latched bounded reaction
Greeting wave	Dynamic	One wrist above its shoulder; two horizontal direction reversals in 1.5 s, each excursion > 0.25 shoulder widths	Left-to-right traveling tilt
Open arms	Dynamic	Wrist separation changes from < 0.6 to > 1.8 shoulder widths within 1.5 s, with both wrists visible	Symmetric outward opening and closing
Raise and lower arms	Dynamic	Both wrists move from below shoulders to above them and back within 2.0 s; each leg exceeds 0.3 shoulder widths	Alternating upper/lower-row oscillation
Peace sign	Static	Index and middle fingers extended (joint angles $> 150^\circ$); ring/little fingers flexed ($< 120^\circ$); hold 0.4 s	Alternating neighboring tilt groups
Two-hand heart	Static	Index-tip pair and thumb-tip pair each within 0.35 mean hand scales; index midpoint above thumb midpoint by > 0.2 scales; inward-facing hands; hold 0.4 s	Repeating wave from centroid toward outer units

error, backlash, calibration, PWM timing, or servo lag. For small independent angular errors, the proposed budget is

$$\sigma_\theta^2 \approx \sigma_{\text{vision}}^2 + \sigma_{\text{mount}}^2 + \sigma_{\text{cal}}^2 + \sigma_{\text{PWM}}^2 + \sigma_{\text{servo}}^2. \quad (15)$$

Correlated terms require covariance terms; systematic bias must be measured separately. Transverse target uncertainty approximately contributes $\delta\theta \simeq \delta p_\perp / d$ radians at target distance d . This explains why numerical precision cannot justify the proposed 5° physical acceptance target without measurement.

G. Actuation, Torque, and Electrical Budgets

For the nominal PCA9685 oscillator $f_{\text{osc}} = 25$ MHz and desired $f_{\text{PWM}} = 50$ Hz [15],

$$P = \text{round}\left(\frac{f_{\text{osc}}}{4096 f_{\text{PWM}}}\right) - 1 = 121. \quad (16)$$

The calculated frequency is 50.0288 Hz and the pulse increment is $4.88 \mu\text{s}$. Frequency must be verified with instrumentation. Calibrate a monotonic per-servo mapping from angle to pulse width rather than assuming a generic 1–2 ms travel range. If the measured slope is $k_t \mu\text{s}$ per degree, one-count angular resolution is approximately $4.88/k_t$ degrees. For illustration only, a 1000 μs span over 180° would give 0.878° per count, not 12-bit resolution over the joint range.

For joint j , a conservative preliminary mechanical requirement is

$$\tau_{j,\text{req}} = \sum_k m_k g_0 \ell_{\perp,jk} + J_{j,\text{eq}} |\ddot{q}_j| + \tau_{j,\text{friction}}. \quad (17)$$

All downstream moving masses belong in the first joint's sum, including the second servo and brackets. Mass is in kg, lever arm in m, and torque in N·m. The quoted 1.8 kgf·cm equals 0.1765 N·m, but stall torque is not a continuous operating rating [16]. A factor-of-two screening margin would limit the calculated requirement to 0.0883 N·m relative to that datum. This is an initial design choice, not a thermal-duty guarantee. Measure mass, center of gravity, current, and sustained-load behavior before retaining the actuator.

For 32 servos the supply condition is

$$I_{\text{supply,rated}} \geq 1.25 \left(\sum_{j=1}^{32} I_{j,\text{peak}} + I_{\text{aux}} \right), \quad (18)$$

where the 25% reserve is provisional and must be reconciled with the actual load simultaneity and temperature. The reference connector is described as 10 A maximum [1]; that statement does not establish a 10 A rating for every installed trace, contact, and wire. Even before derating, 10 A divided by 32 allows only 0.3125 A per servo if all share one path. At a provisional 80% utilization limit the value is 0.25 A, excluding auxiliary loads. As peak current is not established, one connector cannot yet be declared adequate. Separate fused actuator branches and a correctly sized shared return may be necessary. Cascading panels also adds their downstream current to upstream paths.

H. Microtests and Experimental Protocol

Table VI distinguishes executed numerical checks from planned bench tests. The supplied original report remains prior evidence; the current package's Python script is an independent calculation using Eqs. (3)–(6), the defined lattice, and

$$\mathbf{p}(t) = \begin{bmatrix} 180 \cos(2\pi t/T) \\ 130 \sin(2\pi 0.7t/T) \\ 220 + 50 \sin(2\pi 0.5t/T) \end{bmatrix} \text{ mm}, \quad T = 20/12 \text{ s}. \quad (19)$$

Sampling is $t_k = k/30$ s for $k = 0, \dots, 49$. Its 800 directions satisfy the ideal joint interval without unreachable cases; the maximum arccos-based error is 1.21×10^{-6} degrees. The small difference from the prior report is not interpreted as an improvement: origin, sampling, and floating-point evaluation can change such near-zero values. The six boundary cases exercise coincident targets, both poles, a near pole, a target behind the panel, and a forward target.

The software heart-wave screen passes its 2 mm criterion with a minimum gap of 4.25 mm. Sampling 181 configurations is not proof of continuous-time clearance, and the script omits mode-transition geometry and structural parts. It therefore supports keeping the small initial amplitude for further CAD analysis, not unrestricted hardware execution.

Bench work proceeds by increments. First identify the actual board and servo variants and check the rails without a motor load. Then measure PWM timing and calibrate one supported servo, approaching travel limits gradually while observing current. Next test one two-axis mechanism and its geometric envelope, followed by four units and finally the full panel. Use a current-limited supply, oscilloscope/logic analyzer, current sensor or shunt, caliper, mass scale, and external angle reference.

TABLE VI
MICROTEST REGISTER. SOFTWARE OUTPUTS ARE EVIDENCE OF CALCULATIONS ONLY.

ID	Question / test	Decision criterion or observed output	Status
M1	Forward/inverse consistency and boundary handling	800 reachable configurations; max error 1.21×10^{-6} degrees; six boundary cases pass	Executed in supplied Python script
M2	Step-command conditioning	Synthetic 30° step at 30 Hz; largest increment 3.00° ; reaches 95% in 0.33 s	Executed; no motor model
M3	Preliminary heart-wave clearance	181 configurations over 6 s; all 120 unit pairs checked; minimum sphere-envelope gap 4.25 mm	Executed; panels only, no swept-solid proof
M4	Voltage, addressing, pulse generation	No 5 V logic injection; separate driver addresses; oscilloscope verifies approximately 50 Hz and calibrated endpoints	Planned bench microtest
M5	Torque and power scaling	One servo, one unit, four units, then full panel; record startup/load peaks, voltage dips, temperature, and weakest-path rating	Planned; full-panel current unknown
M6	Perception, latching, and ownership	Five gestures including negatives; no heart-phase restart; correct owner under overlap; satisfy R2–R4	Planned camera and state tests
M7	Fault and physical pointing checks	Invalid CRC/sequence, link loss, blocked target, and measured pointing grid; satisfy R5, R7, R8	Planned integration tests

Do not use prolonged intentional stalling to infer continuous performance. Record instrument range, resolution, calibration status, and mounting conditions.

For gesture evaluation, the proposed pilot uses six consenting adult participants: three for threshold development and three held out for evaluation. Each performs five repetitions of each gesture under two recorded illuminance conditions and two distances, initially 1.5 and 2.5 m if camera framing permits. This gives 600 labeled events across both groups, with 300 held-out events. Additional 1-, 2-, and 3-person scenes test ownership and occlusion; for each occupancy and each gesture, collect at least ten held-out-owner trials, plus ten minutes of negative activity per occupancy. Randomize gesture order and keep development and test participants separate. Thresholds are frozen before evaluation.

Report per-class precision, recall, event macro-F1, false activations per minute, processed-frame rate, identity switches, and timing percentiles. A detection is matched to at most one ground-truth event in its annotated completion window plus 0.5 s. Duplicate detections are false positives; unmatched events are misses. Separately measure gesture-completion-to-event, event-to-ACK, and event-to-first-physical-motion delay. Host round-trip timing uses one monotonic clock; cross-device timestamps require an explicit synchronization method. Camera latency cannot be inferred from the PWM refresh frequency.

The acquisition record contains run ID, participant pseudonym, occupancy, lighting, distance, configuration hash, frame time, owner ID, ground-truth event, detected event, confidence, mode, target coordinates, commanded and independently measured angles, sequence/ACK time, rail voltage, current, and fault code. The package supplies empty CSV headers, not fabricated measurements. Raw video is unnecessary for routine operation; retained evaluation clips require consent, controlled access, and a deletion plan. No participant recruitment or human study has been conducted for this report.

I. Risks, Mitigation, and Advancement Decision

Table VII ranks consequences qualitatively because failure probabilities have not been measured. It maps each uncertainty to an action and an observable closure condition rather than assigning an unsupported numerical risk score.

The proposed advancement criterion is documentary completeness plus successful low-power interface and single-unit microtests before full-panel integration. This report provides the reference explanation, opportunity choice, preliminary architecture, analytical model, and software checks. Power capacity, physical torque/clearance, multi-user recognition, and fault response remain empirical gates. Discover–Learn documentation is therefore available for review while those measured feasibility decisions remain explicitly open.

ACKNOWLEDGMENT AND AI ASSISTANCE

The supplied ITESO LabDST teaching material informed the technical scope. OpenAI ChatGPT/Codex assisted with document analysis, literature discovery, English drafting, figure generation, and compilation checks. These activities did not include physical experiments or participant observations. The team remains responsible for technical review, source interpretation, and approval of the final submission.

CONFLICTS OF INTEREST

No conflict declaration was included in the supplied project records. Confirmation by all listed authors is pending before formal submission.

J. Mobile Control Interface

The selected technical proposal includes a mobile interface for remotely controlling the complete sixteen-unit SOFIA panel. The operator will be able to send bounded position commands or activate predefined motion patterns through either a browser-accessible HTML interface or a dedicated mobile application. A smartphone was selected because it is portable, supports wireless communication, and does not require additional user equipment. Although no ITESO-specific ownership survey was found, Mexico’s ENDUTIH 2025 reported that 97.6% of people aged 15–24 used the Internet and that 97.3% of Internet users accessed it through a smartphone, supporting the expected accessibility of this interface among university students [21]. The mobile device will operate only as a supervisory interface: commands will still pass through the ESP32-C3 validation, calibrated angle limits, communication watchdog, and emergency-stop logic before being sent to the servomotors.

TABLE VII
PRELIMINARY RISK REGISTER AND CLOSURE EVIDENCE. ALL LISTED HARDWARE RISKS REMAIN OPEN.

Risk	Impact	Mitigation / design responsibility	Closure evidence
Startup current, common-return drop, or undersized contacts	High	Electrical: measured load budget, protected branches, rated wiring, local decoupling	M5 waveforms and temperature; path rating audit
5 V interface into 3.3 V input; boot-pin conflict	High	Electrical/firmware: verify board wiring; compatible transceiver and pull-ups; boot tests	M4 voltage levels and repeatable startup
Panel/link collision or human pinch point	High	Mechanical: full CAD swept-volume checks, clearance margins, guarded demonstration zone, accessible stop	R5 trajectory clearance and supervised bench inspection
Insufficient torque, heating, or backlash	High	Mechanical/control: measure downstream load; reduce amplitude/acceleration; change servo if needed	Torque budget, endurance test, and measured pointing error
False gesture or identity swap	Medium	Perception: temporal rules, confidence gating, owner lock, negative trials	R2 confusion matrices and occupancy-specific metrics
Stale command or lost communication	High	Firmware: sequence/session checks, heartbeat watchdog, Hold, operator resume	M7 fault injection and timestamped transitions
Simulation geometry differs from assembly	High	Mechanical/control: measure pivots, zero offsets, link lengths, and calibration map	Registered CAD model and external orientation measurements
Latched motion continues unexpectedly	Medium	Interaction: display active pattern; explicit Stop; fault priority; no automatic restart	R3 persistence and stop/resume test
Untraceable versions or undocumented decisions	Medium	Team: package manifest, review log, assigned test records, later repository registration	Team-approved authorship/log and version registration

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